Movie runtime ~ year, analysis

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Are blockbuster movies getting longer? Using top-grossing titles from 1990–2023, this analysis finds runtimes have increased over time. Differences between recent (2013–2023) and older (1990–2000) releases are estimated with bootstrap resampling to capture uncertainty, and a simple linear regression quantifies the trend at about 0.38 minutes per year. Model diagnostics address non-normality and potential autocorrelation, with results robust to these checks. The takeaway is a modest, steady increase—roughly 8–10 minutes over two decades—while noting limits from sample choice and model simplicity.

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1 Introduction

I recently came across an article posing a question:

> Are blockbusters getting (reely) longer?

Fueled by the buzz around Christopher Nolan's *Oppenheimer* being his longest movie (just over 3 hours), I decided to explore this question using publicly available data.

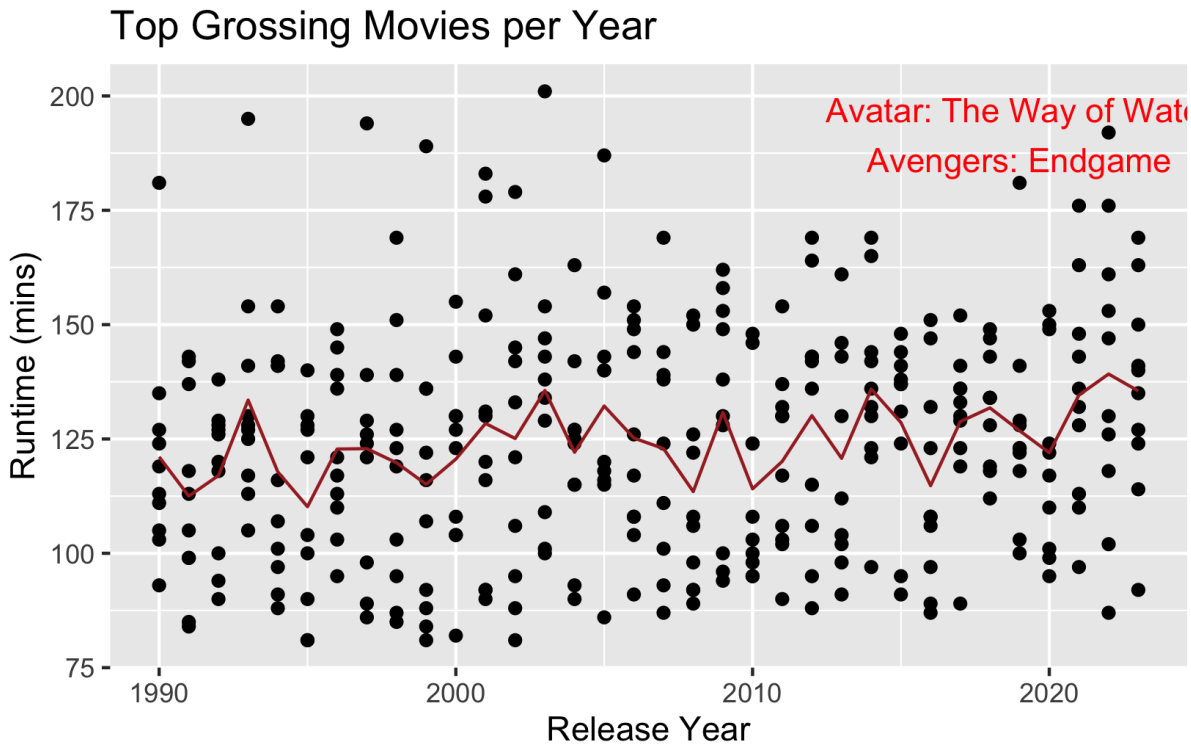
As an exercise in inference this question can be answered with some analysis. I fetched top grossing movies in each year from *boxoffice mojo* and movie details from *imdb* and combined them to create the following dataset.

data

---

2 Data

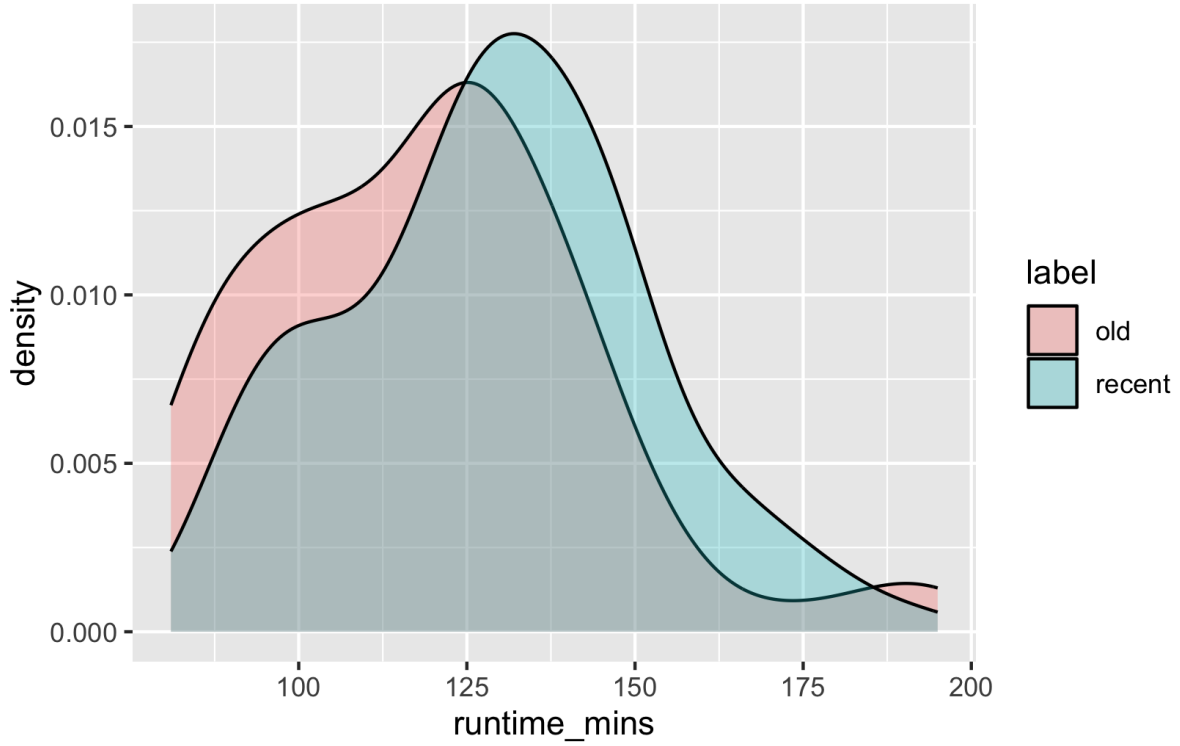
```
# A tibble: 6 x 3
  release_year runtime_mins title
         <dbl>        <dbl> <chr>
1         1990          103 Home Alone
2         1990          127 Ghost
3         1990          181 Dances with Wolves
4         1990          119 Pretty Woman
5         1990           93 Teenage Mutant Ninja Turtles
6         1990          135 The Hunt for Red October
```



3 Analysis

3.1 Comparing recent vs old movies

To simplify the analysis we can consider more recent releases (2013-2023) and compare them to releases from much older timeframe (1990-2000). The runtime distributions for movies in these 2 categories is different although not significantly (visually).



Now we can compare runtimes between two periods. It would have been a straightforward exercise in testing the hypothesis of difference of means using a *t-test* had the distributions of runtimes been more normally distributed.

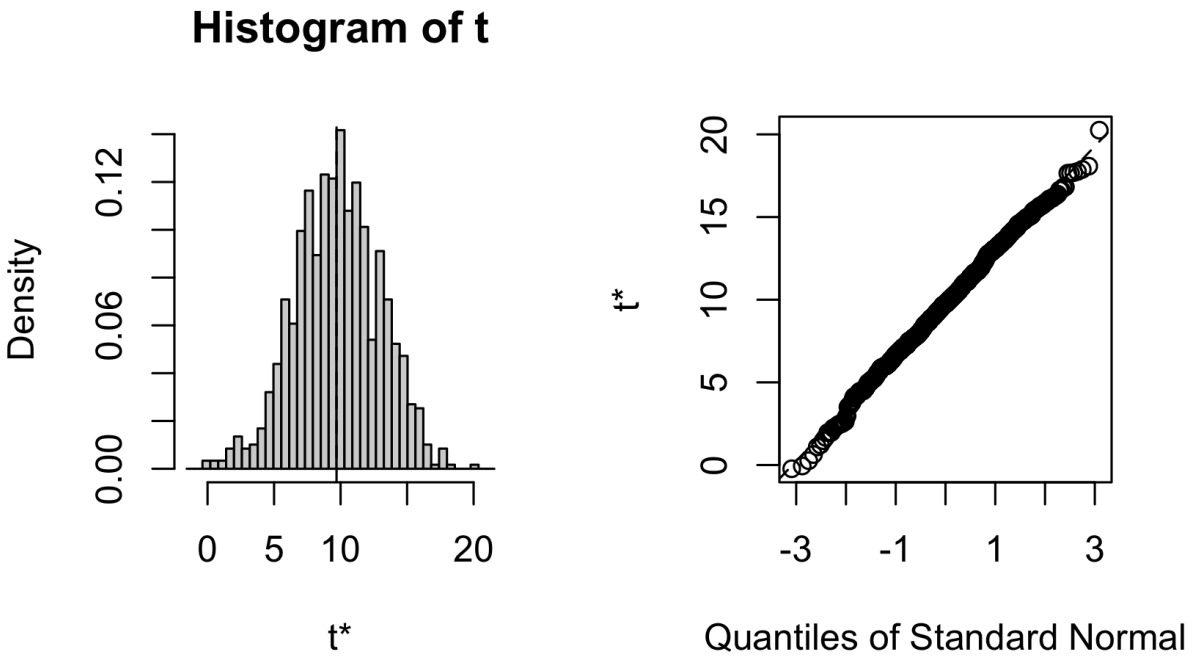
Additionally, the cutoff chosen for *old* (<= 2000) and *recent* (>=2013) are arbitrary. We would ideally want a more comprehensive statement about movie runtimes increasing over the years.

Neverthless, as a preliminary (and simple) analysis we can use bootstrap to compare means of runtimes in the above groups.

```
ORDINARY NONPARAMETRIC BOOTSTRAP


Call:
boot(data = testset, statistic = diff_mean, R = 1000)


Bootstrap Statistics :
    original     bias    std. error
t1* 9.697248 0.06394496    3.174053
```




```
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
Based on 1000 bootstrap replicates

CALL :
boot.ci(boot.out = bootstrap_results, type = "perc")

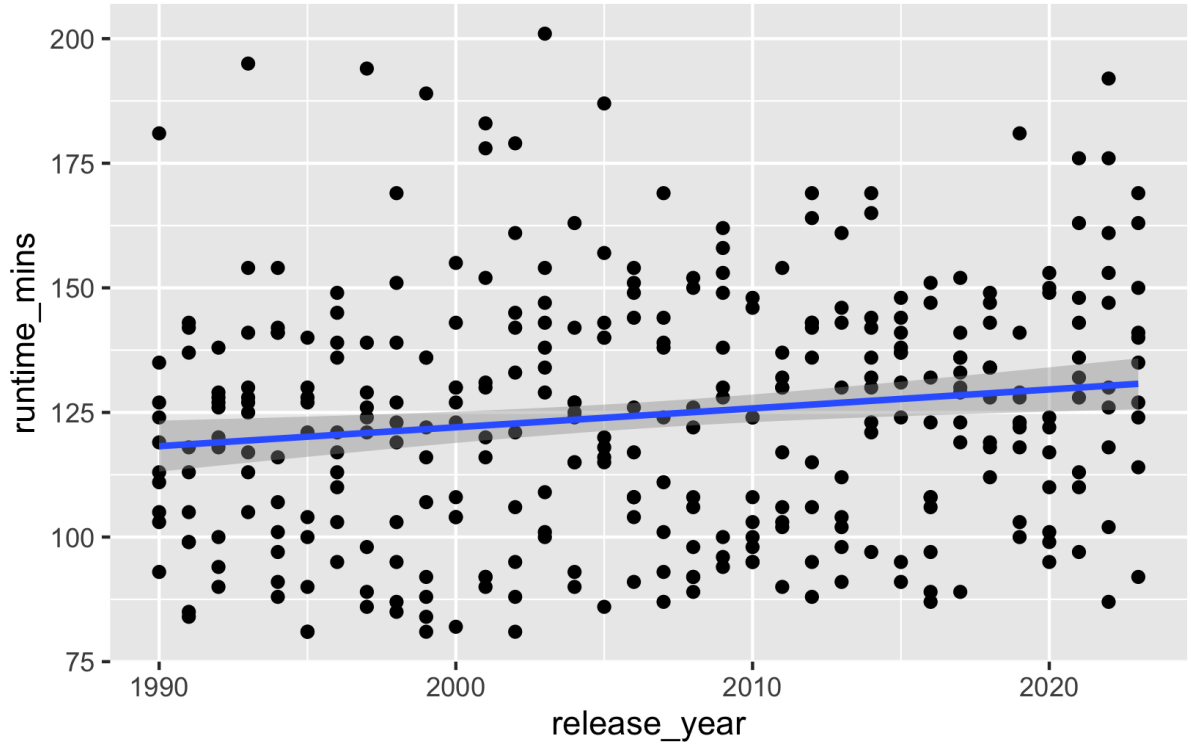
Intervals :
Level     Percentile
95%   ( 3.502, 15.736 )
Calculations and Intervals on Original Scale
```

If we bootstrap the mean of movie times and take the difference between means of recent release years from old, there is evidence that blockbuster movies are getting longer in recent years, and the difference can be up to 10 min on average and can be expected to be between 4 min and 16 min.

So on your next visit to the theatre make sure to get some extra popcorn !!!

3.2 Estimating effect of time

In the want of better precision in identifying the effect of time on movie runtimes, we can formulate a regression problem and estimate the effect of year.



The plot above partly confirms our intuition about the increasing trend of runtime.

```
Call:
lm(formula = runtime_mins ~ release_year, data = yearly_top_movies)

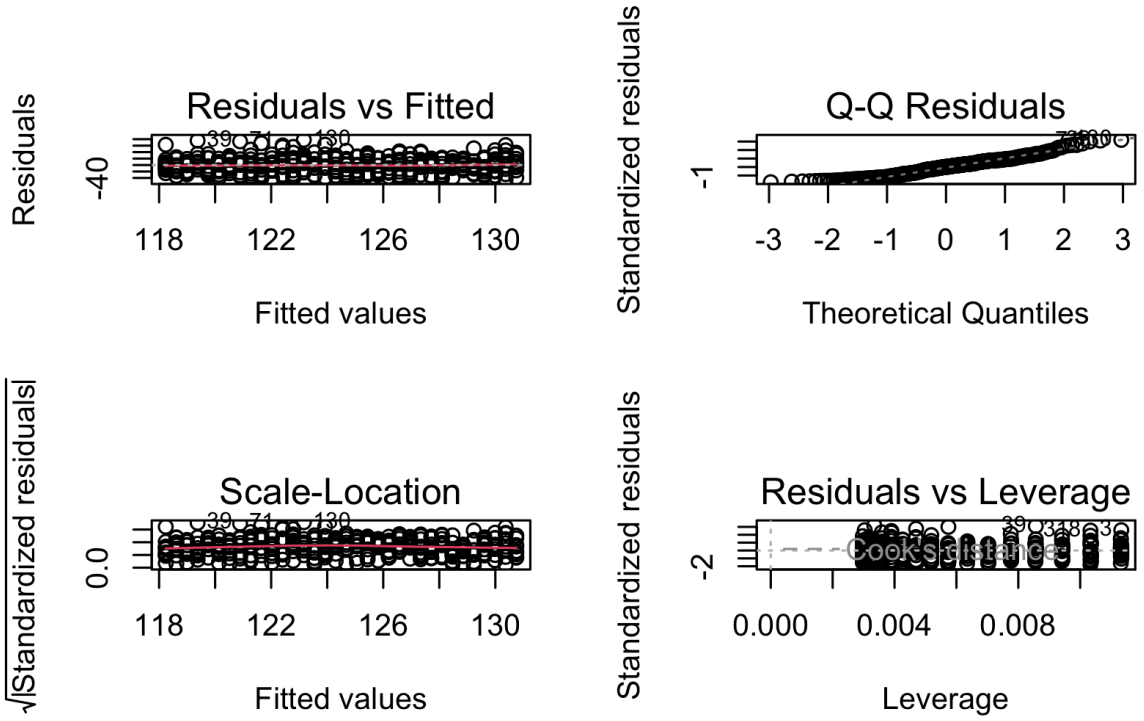
Residuals:
    Min      1Q  Median      3Q     Max
-43.369 -19.617   0.488  16.422  77.834

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -636.1435   271.4520  -2.343  0.01969 *
release_year    0.3791     0.1353   2.802  0.00537 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

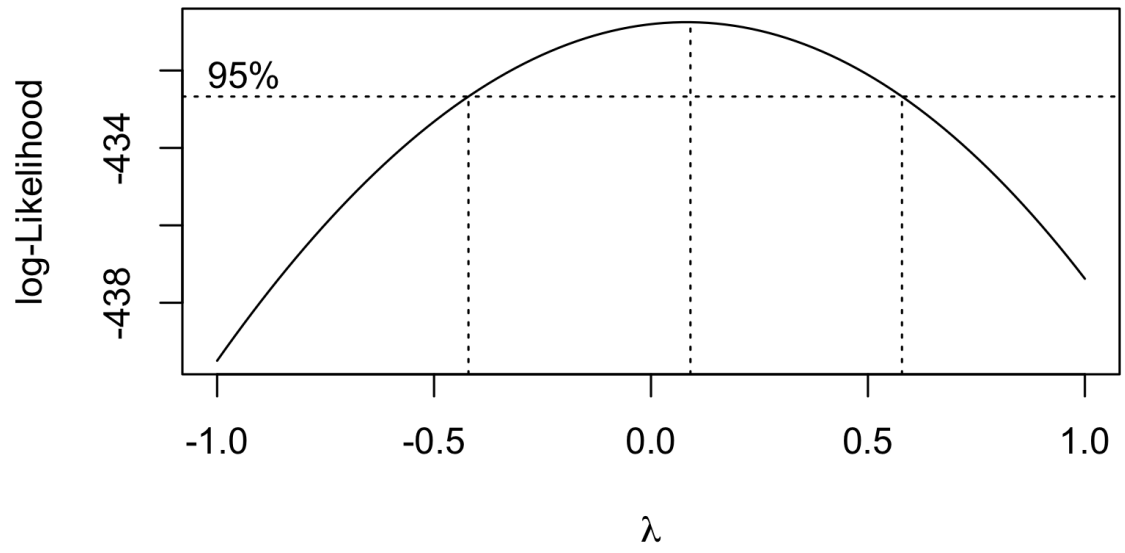
Residual standard error: 24.42 on 335 degrees of freedom
Multiple R-squared:  0.0229,    Adjusted R-squared:  0.01999
F-statistic: 7.852 on 1 and 335 DF,  p-value: 0.005372
```

The regression above supports the hypothesis that *release_year* has positively contributed to movie runtimes at approximately 0.38 min/year. Considering our previous finding on average between 1995 and 2018 the runtime increased by approx. 10 min. From the regression estimate the increase is 0.3791 * (2018-1995) = 8.7 min.

We can certainly run some diagnostics to be sure of our model.



The *residuals* appear to have no pattern with fitted values and also have fairly constant variance. There are no significantly high leverage points or outliers. However, the errors are not quite normally distributed. This can be ignored by relying on large sample size and the fact that other diagnostics are fine. But, we shall check if just in case any transformation of *runtime_mins* may help.



The *Box-Cox* transformation check above hints that there may be some benefit if we used log-transformation (since lambda approximately 0). This thread does not materialise into anything

meaningful as upon transformation, the diagnostics do not change materially and the effect size of *release_year* is practically the same (0.32% per year, 0.38 min/year on base of 119 min). So we will not use any transformation in our model.

There may also be some degree of autocorrelation among residuals due to the fact that we're working with timeseries data. We can test for this autocorrelation and handle it if necessary.

```
Auto-correlation among residuals:  0.01
```

```
    Durbin-Watson test

data:  runtime_mins ~ release_year
DW = 1.982, p-value = 0.4126
alternative hypothesis: true autocorrelation is greater than 0
```

There doesn't appear to be significant autocorrelation and other model diagnotics appear to be fine.

4 Conclusion

There has been a rather steady increase of 0.38 min/year (on average) in movie runtimes as per the dataset we have used. In effect expect to be seated for about 10 more minutes in your favourite blockbuster compared to when you went to movies as a kid.